



*Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences*

Computer Science Department



CS364 Machine Learning

**Term project design, implement, and evaluate
ML models**

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Section : 371**

Group members names	Student IDs
Reema Mohammed Bin Mohsen	441020320
Sarah Abdullah Bagazi	441018657
Fadwa Mohammad Alshumrani	442020917

Instructors: Dr. Waad Alhoshan

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Phase one

1. Introduction

A machine learning model is a mathematical representation of the patterns hidden in data. When the machine learning model is trained (or built or fit) to the training data, it discovers some governing structure within it. That governing structure is formalized into rules, which can be applied to new situations for predictions. So, if we train a model on some training data and then apply that model to new data, the model would be able to infer some relationship within it.

2. Project description

2.1 Main goals

The goal of project is assessment to design, implement, and evaluate various machine learning models (Machine Learning) in a range of real-world applications.

2.2 Objectives

- specify business problem.
- consider risks and success criteria.
- We want to develop our skills in ML and gain more experience.
- We need to learn more about designing and implementing ML models .
- We work in ML model to solve complex real-world problems and make it easy for end users.
- We write the training example to run in model we work on.

2.3 Problem in general

In our project, we will work on a machine learning model that classifies the patients they have covid by them features 1 mean "yes" 2 mean "no" , 97 and 99 mean "missing data "

2.4 Challenges of team

1. At first, the project was not completely clear to us.
2. We had difficulty choosing the project idea.
3. We changed our idea because there was no Dataset for it.
4. We reformulated the document because of the change of idea.
5. We were confined to a short time to modify the idea, so that we searched for other ideas that would have a ready-made dataset.

2.5 Spilt the work of team

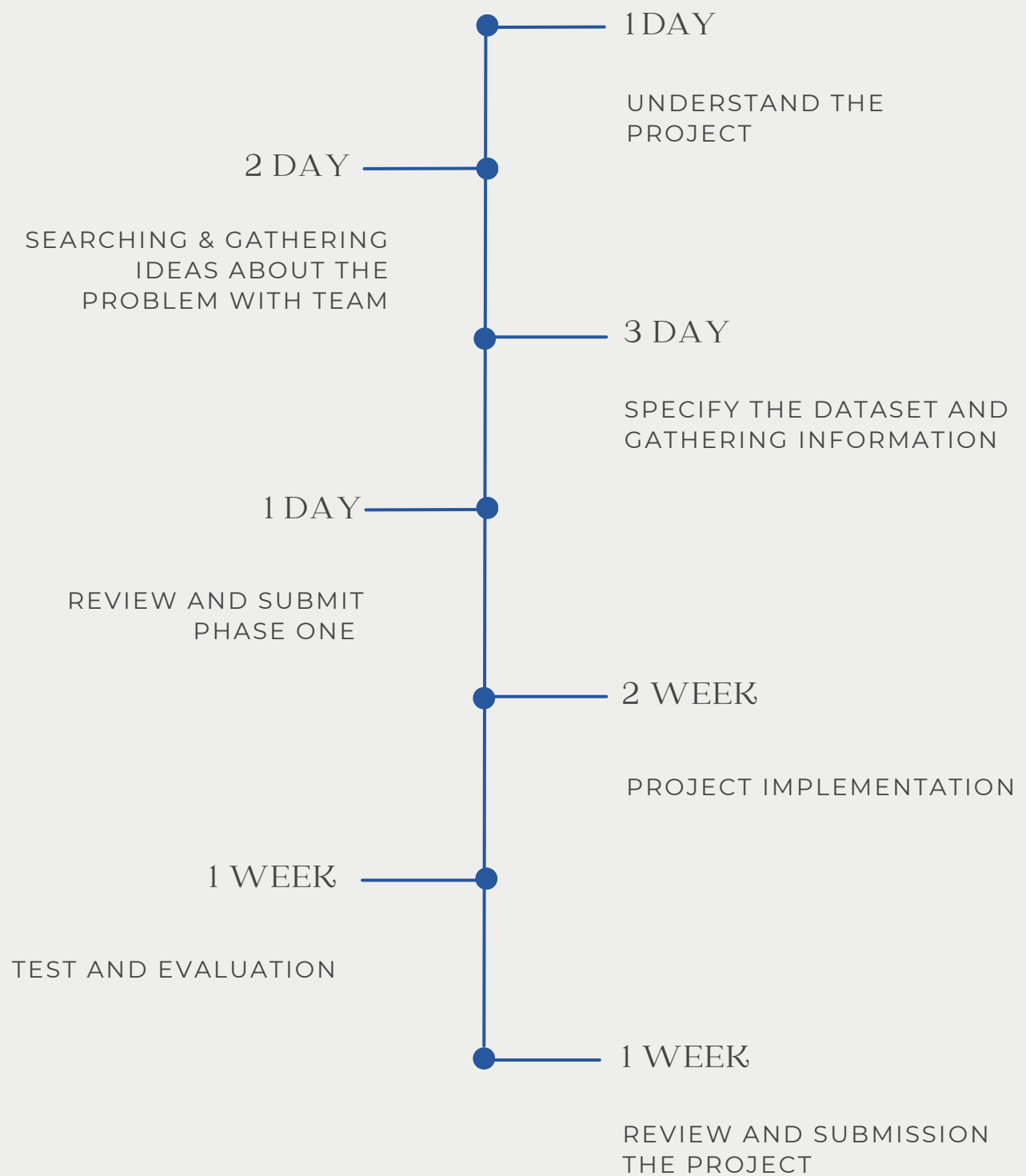
- Reema
 - Designing the cover page.
 - Designing the structure of document.
 - Implementation and programming.
- Sarah
 - Designing the timeline of the Project.
 - Implementation and programming.
 - Reviwe the format.
- Fadwa
 - Testing and evaluation.
 - Implementation and programming.
 - Designing the figures.

2.6 Team members roles and skills

SKILL	REEMA	SARAH	FADWA
Coding	✓	✓	✓
Design			✓
Research	✓	✓	✓
Creative	✓	✓	✓
Hard Worker	✓	✓	✓
Team Management	✓	✓	
Time Managment	✓		
Decision Making	✓	✓	✓
Communication	✓	✓	✓

2.7 Project expected timeline

TIMELINE



3. Project description

3.1 definition of the problem

We identified a problem related to the patients if they infected or not and classify them by features, so that the Machine Learning Model will classify the infected patient by him/his feature so that it is easier for the hospital to recognize the patient info to make doctors refer to patient records so easily

3.2 Type of Machine learning task

The type of the model task is the classification so that when the doctor open the records, Machine Learning model will classifies patient info

3.2.1 why we chose this type in our problem

We choice the classification because the result of the model it will be one of this numbers (1,2,97,99). For example, if we have patient that have covid and after classifying she gets 1 it means pregnant, 2 not pregnant, 97 missing data

4.figures

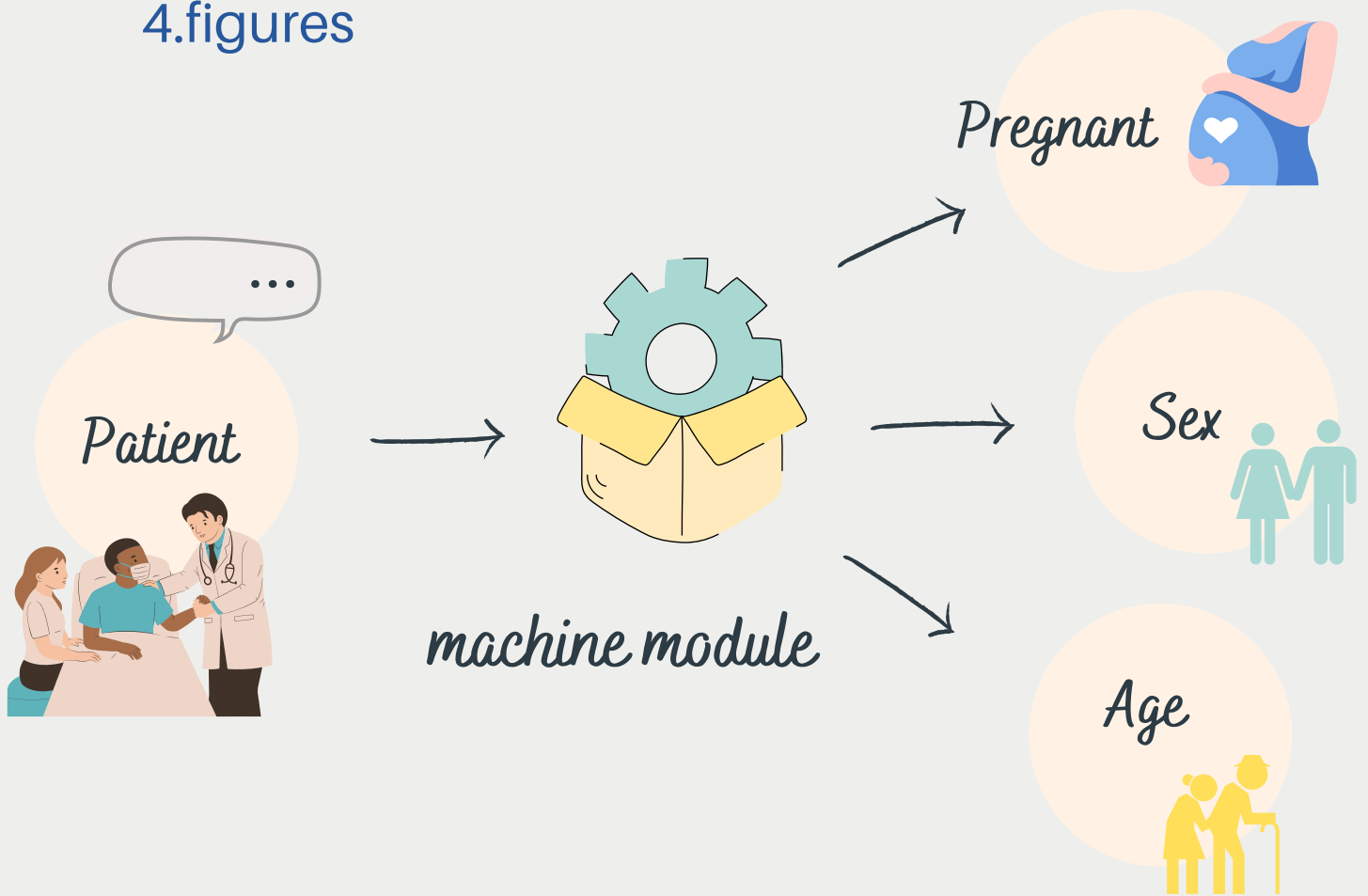


figure1

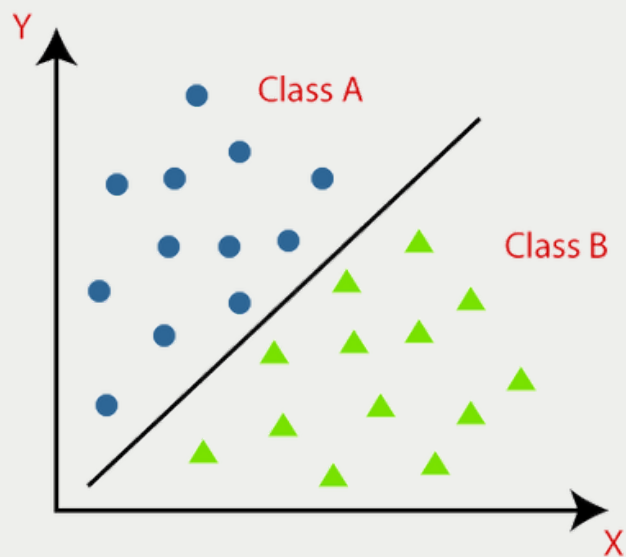


figure 2

5.Dataset description

5.1 Dataset definition

Contains a vast number of anonymized patient-related information including pre-conditions. The raw dataset consists of 21 different features and 1,048,576 unique patients. In the Boolean features, 1 means "yes" and 2 means "no". values as 97 and 99 are missing data

5.2 About Dataset

5.2.1 Context

Coronavirus disease (COVID-19) is an infectious disease caused by a newly discovered coronavirus. Most people infected with COVID-19 virus will experience mild to moderate respiratory illness and recover without requiring special treatment. Older people, and those with underlying medical problems like cardiovascular disease, diabetes, chronic respiratory disease, and cancer are more likely to develop serious illness.

During the entire course of the pandemic, one of the main problems that healthcare providers have faced is the shortage of medical resources and a proper plan to efficiently distribute them. In these tough times, being able to predict what kind of resource an individual might require at the time of being tested positive or even before that will be of immense help to the authorities as they would be able to procure and arrange for the resources necessary to save the life of that patient.

The main goal of this project is to build a machine learning model that, given a Covid-19 patient's current symptom, status, and medical history, will predict whether the patient is in high risk or not.

5.2.2 The attribute and the target

IUSMER : Indicates whether the patient treated medical units of the . first, second or third level.

1 it means patient treated medical units of the first ,2 means second or third level

MEDICAL_UNIT : type of institution of the National Health System that provided the care.Classfie 1 to 13 institution

SEX : 1 mean female. 2 mean male

PATIENT_TYPE : type of care the patient received in the unit.
1 for returned home and 2 for hospitalization.

DATE_DIED : If the patient died indicate the date of death, and 9999-99-99 otherwise.

INTUBED : whether the patient was connected to the ventilator.1 mean connected with device 2 not connected

PNEUMONIA : whether the patient already have air sacs inflammation or not. 1 if he/she have inflammation 2 if not have

AGE : it classifies the age of the patient from 0 to 121

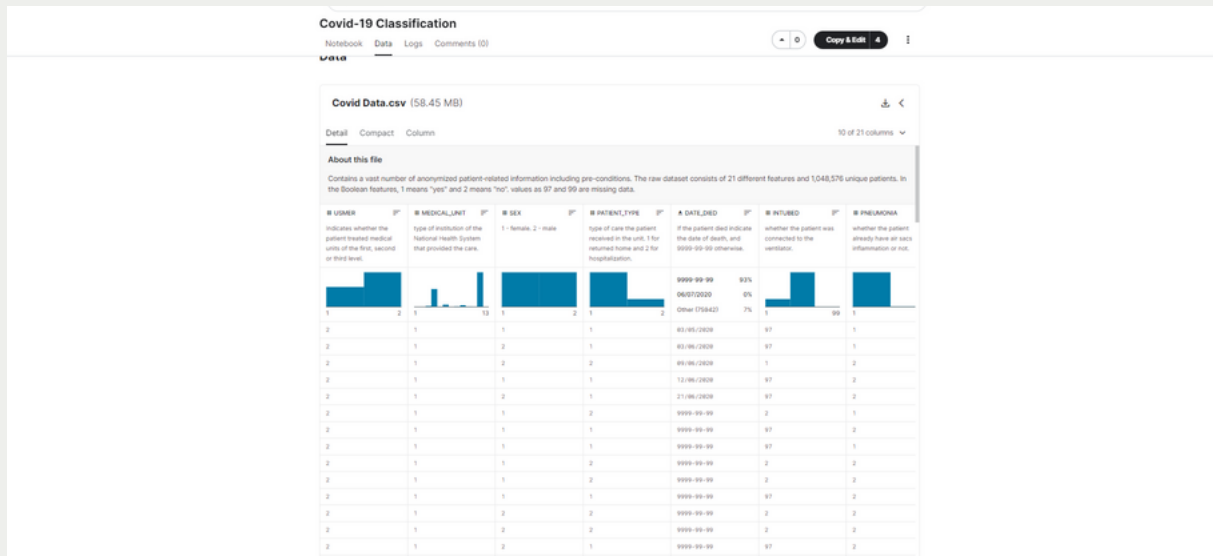
PREGNANT : whether the patient is pregnant or not.1

mean pregnant 2 not pregnant

DIABETES : whether the patient has diabetes or not if
he/she have diabetes gets 1 and 2 if not have .

V	U	I	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
	ICU	CLASIFFR	TOBACC	RENAL	C_OBESITY	CARDIOV	OTHER	CHIPERTEN	INMSUPR	ASTHMA	COPD	DIABETES	PREGNAN	AGE	PNEUMON	INTUBED	DATE_DIE	PATIENT	SEX	MEDICAL	USMER
	97	3	2	2	2	2	2	1	2	2	2	2	2	2	65	1	97	3/5/2020	1	1	1
	97	5	2	1	1	2	2	1	2	2	2	2	2	97	72	1	97	3/6/2020	1	2	1
	2	3	2	2	2	2	2	2	2	2	2	2	1	97	55	2	1	9/6/2020	2	2	1
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figure 3



covid-19 dataset in detail
figure 4

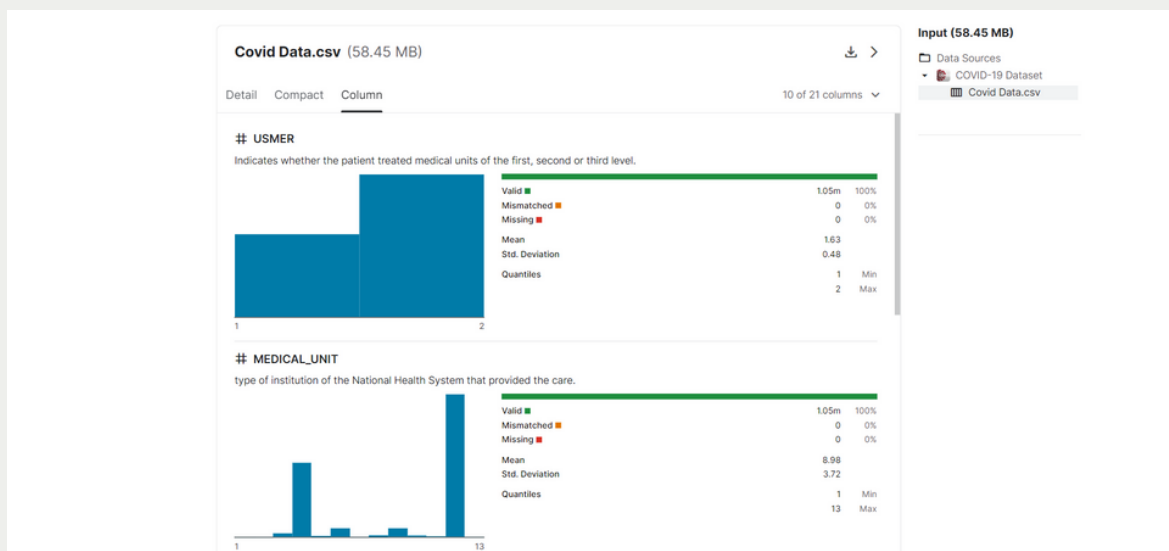


figure 5

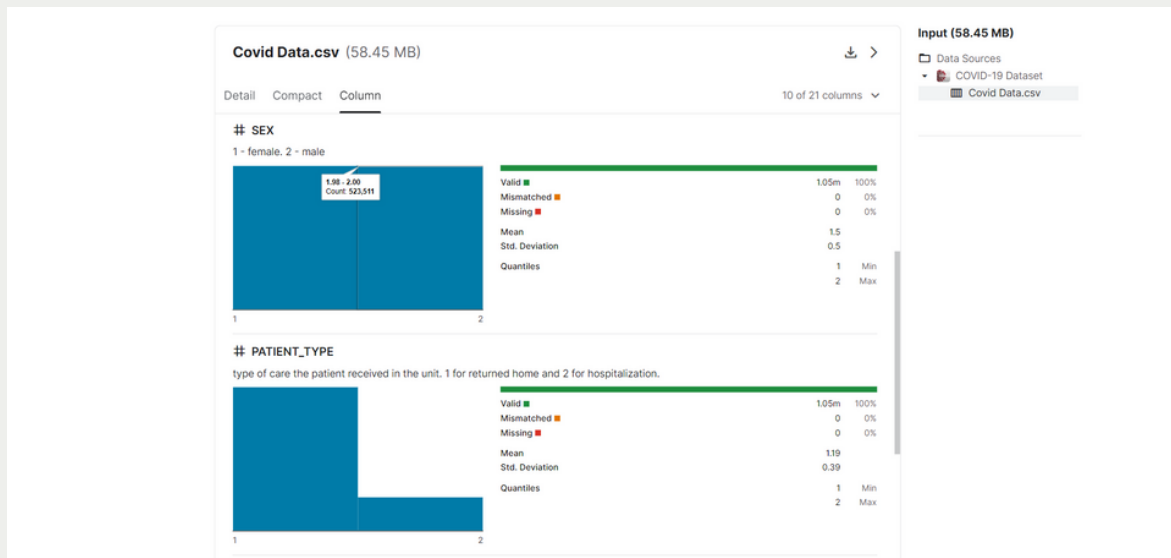


figure 6

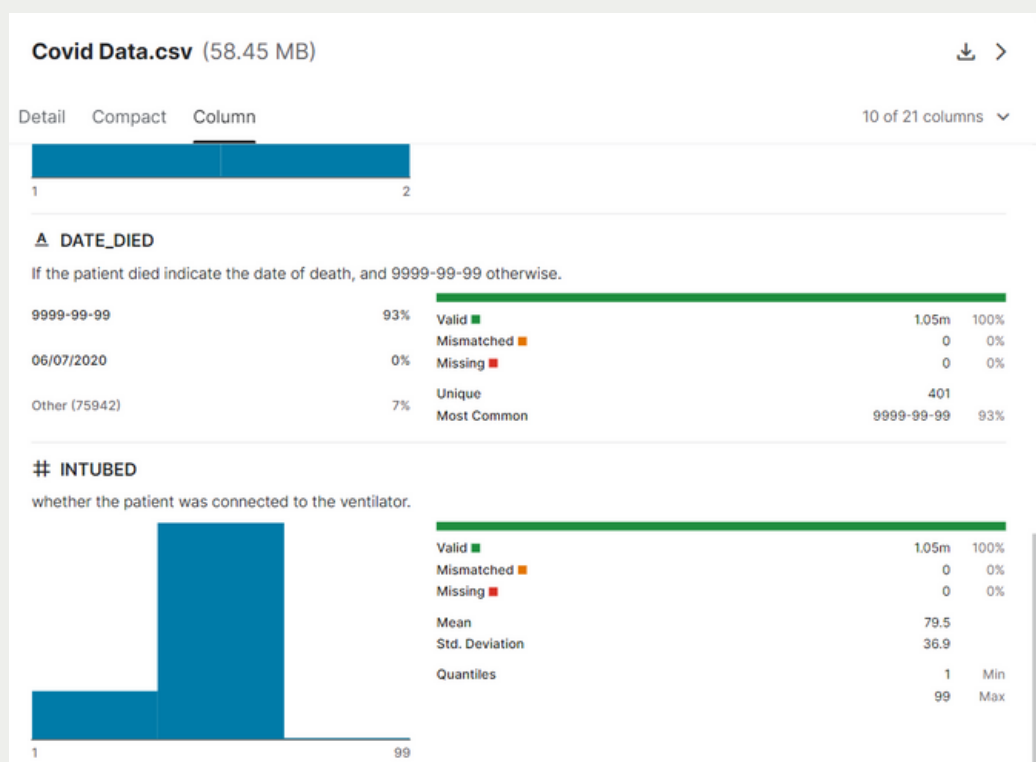


figure 7

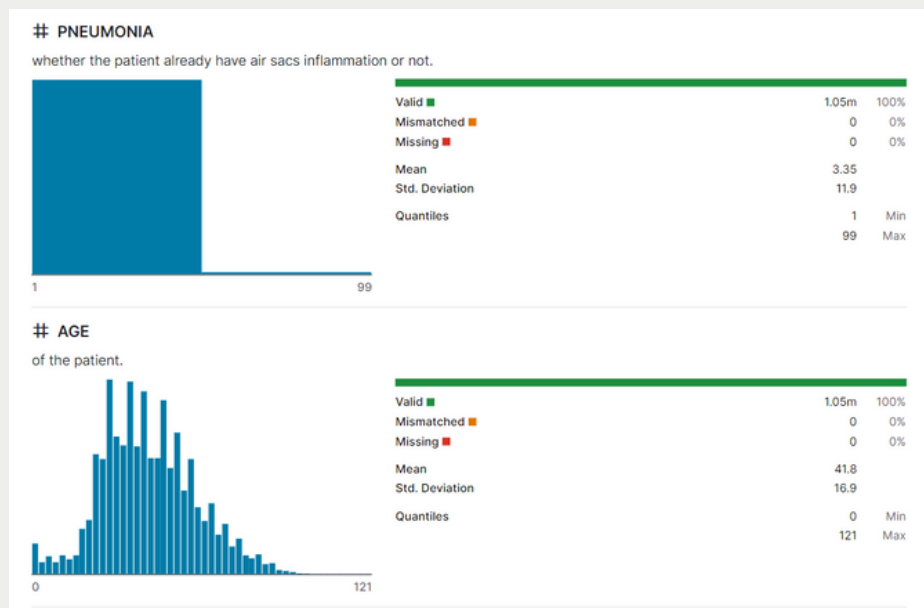


figure 8

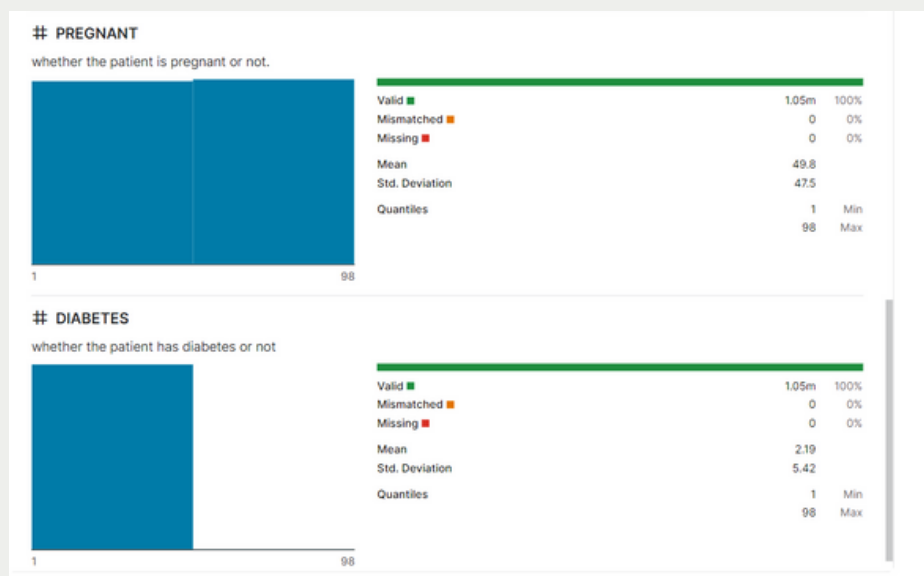


figure 9

6. Machine Learning Algorithm

6.1 What is the algorithm used

We choose K-Nearest Neighbor and logistic regression because they are one of the types of classification we have studied, and we expect it to be relevant to our problem.

K-Nearest Neighbor

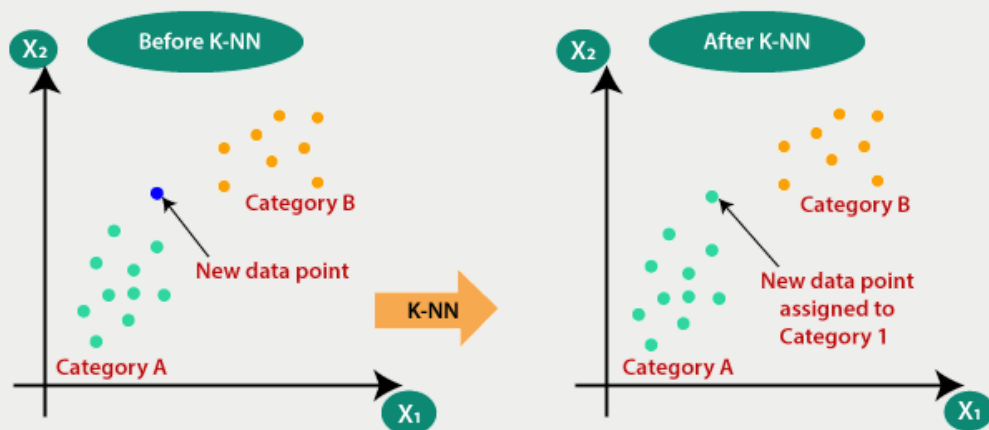


figure 10

logistic regression

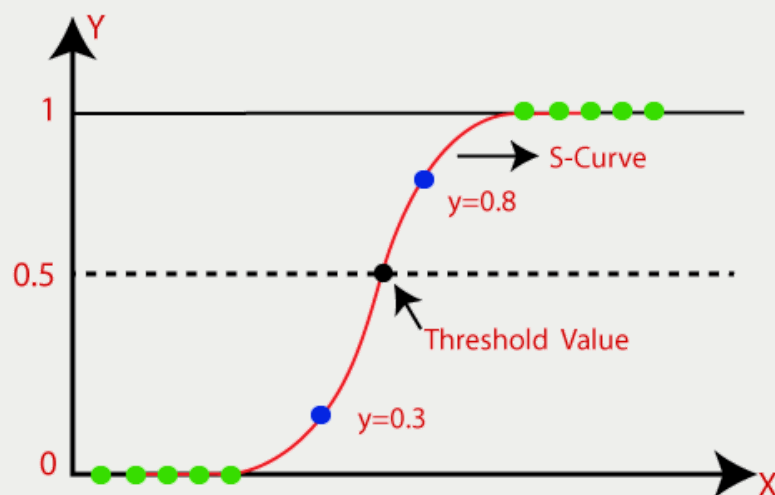


figure11

6.2 K-Nearest Neighbor

- K-Nearest Neighbour is one of the simplest Machine Learning algorithms based on Supervised Learning technique.
- K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories.
- K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm.
- K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.
- K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data.
- It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.
- KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

6.3 Example of K-NN

Example: Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog. So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.

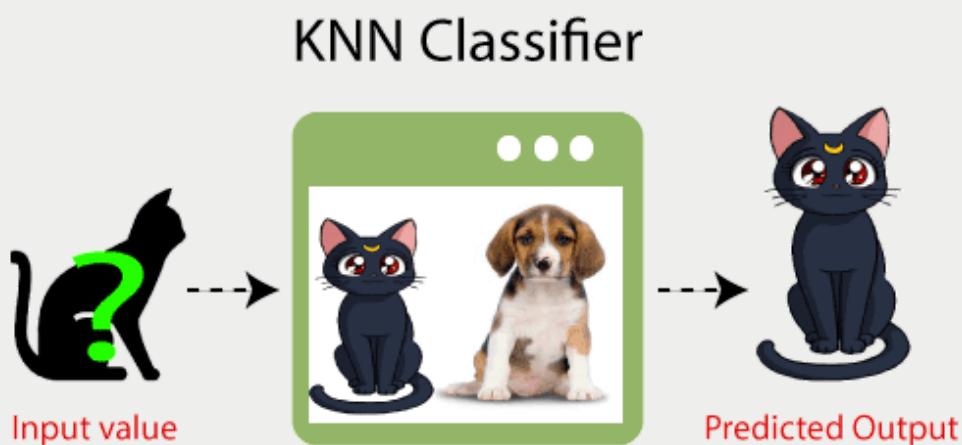


figure 12

6.4 Logistic regression

- Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables.
- Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.
- Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas Logistic regression is used for solving the classification problems.
- In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).
- The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not based on its weight, etc.
- Logistic Regression is a significant machine learning algorithm because it has the ability to provide probabilities and classify new data using continuous and discrete datasets.
- Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification.

6.5 Example of logistic regression

Example: There is a dataset given which contains the information of various users obtained from the social networking sites. There is a car making company that has recently launched a new SUV car. So the company wanted to check how many users from the dataset, wants to purchase the car.

For this problem, we will build a Machine Learning model using the Logistic regression algorithm. The dataset is shown in the below image. In this problem, we will predict the purchased variable (Dependent Variable) by using age and salary (Independent variables).

User ID	Gender	Age	EstimatedSalary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0
15728773	Male	27	58000	0
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	0
15727311	Female	35	65000	0
15570769	Female	26	80000	0
15606274	Female	26	52000	0
15746139	Male	20	86000	0
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1
15617482	Male	45	26000	1
15704583	Male	46	28000	1
15621083	Female	48	29000	1
15649487	Male	45	22000	1
15736760	Female	47	49000	1

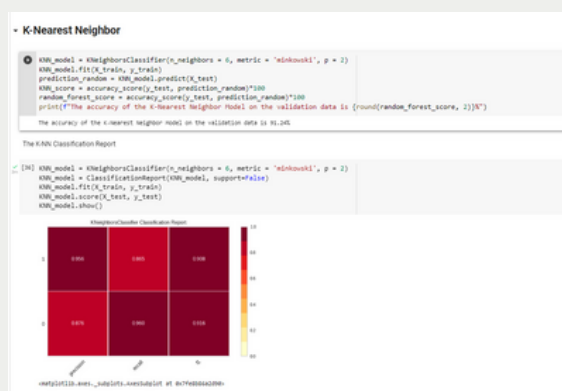
figure 13

7. Model Training & Testing

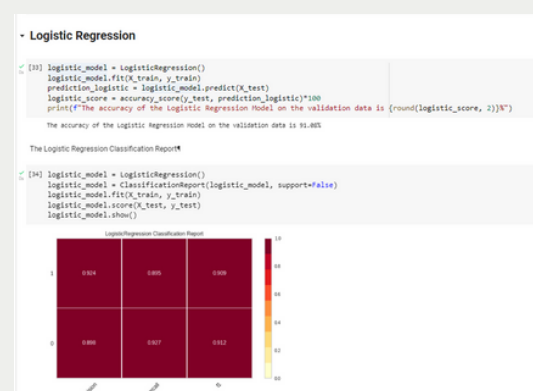
This example of our machine learning model and how it works as shown above for example we will put info about patient that have Covid-19 and the results of the classification will be the patient is female and her age 29, pregnant and she didn't have Diabetes or pneumonia. We split the dataset into training and testing to test the accuracy to each of the both algorithms. After testing it, became clear to us that Logistics is better than K-NN in terms of accuracy.

8. Results & Discussion

After analyzing and comparing between K-NN and Logistic regression algorithm the result of the accuracy of k-nearest neighbor model on the validation data is 90.11%. and the accuracy of logistic regression model on the validation data is 91.08%. So, the logistic regression Model by means of the accuracy on the validation data and its classification report is the best model.



K-NN algorithm



Logistic regression algorithm

9.conclusion

There are many subtleties and pitfalls in ML, and many ways to be lead astray by what appears to be a perfectly well-tuned thinking machine. Almost every part of the basic theory can be played with and altered endlessly, and the results are often fascinating. Many grow into whole new fields of study that are better suited to particular problems.

Machine Learning is an incredibly powerful tool. In the coming years, it promises to help solve some of our most pressing problems, as well as open up whole new worlds of opportunity. The demand for Machine Learning engineers is only going to continue to grow, offering incredible chances to be a part of something big.

In this report, we presented our work on machine learning model and we choose dataset from Kaggle and that dataset classified patient coved-19 based on there attributes for each patient , the type task that suit dataset is classification to classify discrete values and we choose K-NN algorithm and based on that we write the code and during our use to the model We have satisfying results.

10.References

Dataset Covid-19 Classification :

<https://www.kaggle.com/code/tsaqifmu/covid-19-classification/notebook>

K-NN Algorithm :

<https://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-learning>

figures :

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Logistic Regression algorithm :

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